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The Loyola Index Family and the Resolution Limit of Degree Imbalance

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Abstract

Vertex-degree-based indices cannot distinguish graphs with equal edge-degree-pair counts, and indices that depend on each edge only through its degree ratio lose further information. We determine where this further loss begins for trees. The *Loyola index family* $LO(G;\alpha,\beta,\gamma)$ multiplies the edge term $(d_u d_v)^\alpha (d_u + d_v)^\beta$ of the Gutman–Milovanović index by $e^{\gamma q_{uv}}$, where $q_{uv} = |d_u - d_v|/(d_u + d_v)$ is the normalised degree imbalance of the edge. By theorems of Rada, the points $LO(G;0,0,\gamma)$ at algebraic $\gamma \neq 0$ separate exactly the histograms of edge imbalances, and no degree-ratio index, such as the geometric–arithmetic or degree-ratio Sombor index, separates more. By exhaustive enumeration, this histogram determines the degree-pair counts of every tree up to order twelve and of every chemical tree, the carbon skeleton of an alkane, up to order fifteen. Explicit counterexamples exist at every larger order. At fixed (α,β), two chemical graphs exchange order at most five times as γ varies. Pre-registered tests on measured alkane boiling points did not demonstrate a predictive gain from γ. A vaporisation-enthalpy gain that met a pre-registered

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seed-count rule shrank to about zero in an exploratory rerun with a search box that did not truncate the parent index.

1 Introduction

Every vertex-degree-based index depends on a graph only through its edge-degree-pair counts, so graphs with equal counts are never separated [38]. On graphs with $\Delta \leq 4$ every bond-incident-degree (BID) index is a linear functional of the ten edge-degree-pair counts, a representation used in [37, 39] and developed systematically in [38]. An index can lose further information through the form of its edge weights. If the weight of an edge depends only on the ratio of its end degrees, the index sees only the histogram of these ratios. This paper asks at which tree orders that second loss begins, and what a parameter for the degree imbalance adds inside the space of counts.

Degree-based topological indices form one of the most developed descriptor classes of chemical graph theory. The class runs from the Randić and Zagreb indices [23, 41] and their parametric extensions, the general Randić and sum-connectivity families [8, 54], to the Sombor index [17] and its many variants, surveyed in [18]. Extremal results on BID indices are surveyed in [2], and a finite-dimensional linear framework for such indices is developed in [38].

The *Gutman–Milovanović family* [20], studied in the recent bounds and QSPR literature [16, 31, 36], is $M_{\alpha,\beta}(G) = \sum_{uv\in E(G)}(d_u d_v)^{\alpha}(d_u + d_v)^{\beta}$, $\alpha, \beta \in \mathbb{R}$. Its special cases, catalogued in [31], include $M_{0,1} = M_1$, $M_{1,0} = M_2$, the hyper-Zagreb index $M_{0,2} = HM$ [46], $M_{-1,0} = {}^{m}M_2$, the Randić index $M_{-1/2,0} = R$, the sum-connectivity index $M_{0,-1/2} = \chi$ [53], the harmonic index $2M_{0,-1} = H$, the inverse sum indeg index $M_{1,-1} = ISI$, the geometric–arithmetic index $2M_{1/2,-1} = GA$ [49] and the arithmetic–geometric index $\frac{1}{2}M_{-1/2,1} = AG$. Its two parameter axes are the general Randić and sum-connectivity families, and parameter-optimised versions have already been fitted to octane properties [3].

We study the three-parameter extension

$$LO(G;\alpha,\beta,\gamma) = \sum_{uv\in E(G)} (d_u d_v)^{\alpha}(d_u+d_v)^{\beta} e^{\gamma q_{uv}}, \qquad q_{uv} = \frac{|d_u - d_v|}{d_u + d_v},$$

the *Loyola index family*, with $LO(G;\alpha,\beta,0) = M_{\alpha,\beta}(G)$. Positive γ upweights degree-imbalanced edges, negative γ downweights them, and on regular graphs γ has no effect. We ask what γ adds. It reads no graph information beyond the degree-pair counts, but at rational parent exponents it can separate graphs that the parent merges.

The family is a three-parameter subfamily of the exponential vertex-degree-based indices of Rada [37], whose extremal theory is developed in [9, 15, 47] and, for general edge weights, in [24]. Published exponential indices [11, 12] exponentiate a single classical generator. The variable Euler–Sombor index [22] and the cosine-rule Sombor indices [34] weight the imbalance polynomially and without normalisation. Functions of the ratio q_{uv} appear in the diminished Sombor index [32], in degree-ratio indices [19,35], and, as a normalised form of Albertson's irregularity [1,28], in $IRLA(G) = 2\sum_{uv} q_{uv}$ [43]. The pure-γ point $LO(G;0,0,\gamma) = \sum_{uv} e^{\gamma q_{uv}}$ is the exponential index, in the sense of [37], of the edge function γq_{uv}. The coordinate added here is linear in q_{uv} in the exponent and is coupled with its own parameter to the Gutman–Milovanović family. It makes every edge weight log-linear in (α,β,γ), so LO is the partition function of an exponential family on the edges of G.

For $\Delta \le 4$ the whole family lies in the fixed ten-dimensional space of linear functionals of the counts. Our questions concern a nonlinear family inside that space: which count vectors a parameter point separates, and how the index moves along γ. The tools are classical. Gutman's survey [18] observes that new variants hardly ever improve on earlier ones in applications, and our tests agree. The family is not proposed as a better predictor, and the QSPR sets used here are small.

What is new here is where the known results bind for chemical graphs. By exhaustive enumeration, the q-histogram carries the full degree-pair information of an alkane carbon skeleton up to order 15 and not beyond, so no degree-ratio index, such as GA, AG or the diminished Sombor index,

separates all skeletons of order 16 or more. At fixed (α, β) the order of two chemical graphs changes at most five times along γ.

The results below differ in status, and we group them accordingly.

(i) *Known theorems applied to the family.* That the pure-γ points at algebraic $\gamma \neq 0$ separate exactly the q-histograms is, for graphs of equal order, Theorem 3.4 of [37] applied to the weights γq_{ij}, and that no degree-ratio index separates more is Proposition 2.5 there (Theorem 2(ii) and the corollary on degree-ratio indices in Section 3.1). The same Lindemann–Weierstrass argument, with algebraic coefficients, shows that at rational exponents a nonzero algebraic γ never merges two graphs that the parent separates (Proposition 3). The extremal trees of the pure-γ points follow from the criteria of Cruz and Rada [10] (Section 5). Two octane collision pairs realise the known degree-pair ceiling [38, Section 3.1] (Proposition 5).

(ii) *Results proved here.* Explicit trees with distinct profiles and equal q-histograms at every order from 13, and from 16 for $\Delta \leq 4$ (Theorem 2(iii)), carried by the corollary to every degree-ratio index. A bound on the number of order exchanges along γ (Proposition 4), an application of Laguerre's rule of signs, and per-graph identifiability, stated and proved in Supplementary Section S10. Two elementary facts: the imbalance direction is affinely independent of the parent coordinates once a vertex of degree at least 3 is admitted (Lemma 1), and $\log LO$ is a log-sum-exp and hence convex (Theorem 1).

(iii) *Computer-assisted results.* That these orders are the first is established by exhaustive enumeration, as are the value counts for trees of orders 7 to 17 (Section 4).

(iv) *A diagnostic application.* Structure sensitivity on the decanes (Section 4.3), a matched-budget QSPR diagnostic and pre-registered tests on measured boiling points (Section 6). Freeing γ gives no demonstrated gain once the parent's search box does not truncate it (Section 6.2). On measured boiling points ridge regression on the degree-pair counts leads every single index evaluated in point estimate (Section 6.4).

2 Preliminaries and notation

Let $G = (V, E)$ be a finite simple connected graph with at least one edge, of order $n = |V|$ and size $m = |E|$, and for a vertex $v \in V$ let the degree $d_v = d_G(v)$ count the edges incident to v. The maximum and minimum degrees are denoted $\Delta = \Delta(G)$ and $\delta = \delta(G)$. The chemometric sections use hydrogen-suppressed molecular graphs of alkane carbon skeletons, for which $\Delta \le 4$, while the mathematical results do not assume this bound unless stated.

For an edge $e = uv \in E(G)$ we write (d_u, d_v) for its *degree pair*, understood as an unordered multiset $\{d_u, d_v\}$. For integers $1 \le i \le j$ let $m_{ij}(G) = |\{uv \in E(G) : \{d_u, d_v\} = \{i, j\}\}|$ be the number of edges with degree pair (i, j), called (i, j)-edges; the vector $(m_{ij}(G))_{i \le j}$ is the *degree-pair profile* (or count vector) of G.

Definition 1 (Loyola index family)**.** Let $G = (V, E)$ be a simple connected graph. For $\alpha, \beta, \gamma \in \mathbb{R}$, the *Loyola index family* is defined as

$$LO(G; \alpha, \beta, \gamma) \;=\; \sum_{uv \in E(G)} (d_u d_v)^{\alpha} \, (d_u + d_v)^{\beta} \, \exp\!\left(\gamma \, \frac{|d_u - d_v|}{d_u + d_v} \right). \tag{1}$$

Eq. (1) is defined for all $\alpha, \beta, \gamma \in \mathbb{R}$ on graphs without isolated vertices. Write $q_{uv} = \mathrm{imb}(d_u, d_v)$, where $\mathrm{imb}(x, y) = |x - y|/(x + y)$, for the normalised degree imbalance of the edge uv, and $q_{ij} = (j - i)/(i + j)$ for $i \le j$. Each summand depends only on the unordered degree pair of its edge, so

$$LO(G; \alpha, \beta, \gamma) = \sum_{i \le j} m_{ij}(G) \, (ij)^{\alpha} (i + j)^{\beta} e^{\gamma q_{ij}}. \tag{2}$$

In particular $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$, the Gutman–Milovanović family [20, 31], and $LO(G; \alpha, \beta, \gamma) = M_{\alpha,\beta}(G)$ for every γ when G is regular.

Definition 2 (q-histogram)**.** The q-histogram of G is the function $n_q(G) = \sum_{i \le j,\; q_{ij} = q} m_{ij}(G)$, $q \in [0, 1)$, the number of edges of G with imbalance q.

Equal profiles give equal q-histograms. The converse fails (Theorem 2(iii)).

3 Parameter geometry

Throughout this section $\theta = (\alpha, \beta, \gamma)$ and $z_{uv} = \big(\ln(d_u d_v), \ln(d_u + d_v), q_{uv}\big)$, so that

$$LO_G(\theta) = \sum_{uv \in E(G)} e^{\theta^\top z_{uv}}. \tag{3}$$

For a degree pair (i, j) we write z_{ij}, and call $(1, z_{ij}) \in \mathbb{R}^4$ its augmented feature vector. A set of augmented vectors spans $\mathbb{R}^4$ if and only if the corresponding z_{ij} do not lie in one affine plane of $\mathbb{R}^3$.

Lemma 1 (Affine independence of the imbalance direction)**.** *For $\Delta \geq 3$, with $P_\Delta = \{(i,j) : 1 \leq i \leq j \leq \Delta\}$ the admissible degree pairs, the augmented feature vectors $\big(1, \ln(ij), \ln(i+j), q_{ij}\big)$, $(i,j) \in P_\Delta$, span $\mathbb{R}^4$. Consequently, if two parameter triples θ, θ' satisfy $e^{\theta^\top z} = c\, e^{\theta'^\top z}$ for some constant $c > 0$ and all $(i,j) \in P_\Delta$, then $\theta = \theta'$ and $c = 1$. In particular, no choice of (α', β') and scaling constant reproduces the action of $\gamma \neq 0$ on all admissible degree pairs. For $\Delta = 2$ the span has dimension 3, while for $\Delta = 1$ the only pair is $(1,1)$, with augmented vector $(1, 0, \ln 2, 0)$, and the span has dimension 1. In both cases the conclusion fails. For $\Delta = 2$, indeed, $q_{ij} = [\ln(i+j) - \ln 2 - \frac{1}{2}\ln(ij)]/(3\ln 3 - \frac{9}{2}\ln 2)$ on the three pairs, so for $\Delta = 2$ the factor $e^{\gamma q_{ij}}$ equals a constant times $(ij)^a (i+j)^b$ on the three pairs, for suitable constants a, b. The effect of γ is therefore reproduced exactly by a shift of (α, β) together with a scaling constant.*

Proof. The vectors $(1, \ln(ij), \ln(i+j))$ on the pairs $(1,1), (2,2), (1,2)$ have rank 3, since the corresponding 3×3 determinant equals $\ln 2 \cdot \ln(9/8) \neq 0$, so it suffices to show that q_{ij} is not an affine function of $\big(\ln(ij), \ln(i+j)\big)$ on P_Δ. The four pairs $(1,1), (2,2), (1,2), (1,3)$, all admissible once $\Delta \geq 3$, witness this. The pair $(1,1)$ occurs only in K_2, and the conclusion persists without it, since the augmented vectors of $(1,2), (1,3), (2,2), (3,3)$, namely $(1, \ln 2, \ln 3, \frac{1}{3})$, $(1, \ln 3, \ln 4, \frac{1}{2})$, $(1, \ln 4, \ln 4, 0)$ and $(1, \ln 9, \ln 6, 0)$, have determinant $-\frac{1}{6}\ln\frac{3}{2}\,\ln\big(2^{13}/3^8\big) \approx -0.0150 \neq 0$. Suppose $q_{ij} = a + b\ln(ij) + c\ln(i+j)$ on these pairs. The pairs $(1,1)$ and $(2,2)$ have $q = 0$, so they give $a + c\ln 2 = 0$ and $a + (2b + 2c)\ln 2 = 0$, hence $a = -c\ln 2$ and $b = -c/2$. Substituting into the $(1,2)$ and $(1,3)$ equations gives $\frac{1}{3} = c\,(\ln 3 - \frac{3}{2}\ln 2)$ and $\frac{1}{2} = c\,(\ln 2 - \frac{1}{2}\ln 3)$. In particular $c \neq 0$, since the

left side $\frac{1}{3}$ is nonzero. Three times the first minus twice the second gives $0 = c\,(4\ln 3 - \frac{13}{2}\ln 2)$, hence $8\ln 3 = 13\ln 2$, i.e. $3^8 = 2^{13}$. This is false ($6561 \neq 8192$). Hence the augmented vectors have rank 4. Indeed, a linear relation among the four coordinates on P_Δ with a nonzero coefficient on q_{ij} would express q_{ij} as an affine function of $\ln(ij)$ and $\ln(i{+}j)$, which we have just excluded. A relation with zero coefficient on q_{ij} is excluded by the rank-3 determinant. If $e^{\theta^\top z} = c\,e^{\theta'^\top z}$ on P_Δ, then $(\theta-\theta')^\top z_{ij} = \ln c$ for all admissible pairs, so $(\ln c, \theta' - \theta)$ is orthogonal to every augmented vector. Since the augmented vectors span $\mathbb{R}^4$, the only vector orthogonal to all of them is 0, so $\theta = \theta'$ and $c = 1$. For $\Delta = 2$ only the pairs $(1,1), (1,2), (2,2)$ exist. The 3×3 determinant computed at the start of the proof shows that their three augmented vectors have rank 3, and three vectors cannot have more. For $\Delta = 1$ the single augmented vector has rank 1. ■

By (3), $\log LO_G$ is a log-sum-exp function of θ.

Theorem 1 (Log-convexity)**.** *Fix a graph G with $m \geq 1$ edges, define $F_G(\theta) = \log LO_G(\theta)$ on $\mathbb{R}^3$, and let $\mathbb{E}_\theta$ and Cov_θ denote mean and covariance under the edge probability measure $w_{uv}(\theta) = e^{\theta^\top z_{uv}}/LO_G(\theta)$. Then F_G is convex, with*

$$\nabla F_G(\theta) = \mathbb{E}_\theta[z], \qquad \nabla^2 F_G(\theta) = \mathrm{Cov}_\theta(z) \succeq 0.$$

In particular $\partial_\gamma F_G = \mathbb{E}_\theta[q] \in [0,1)$ and $\partial_\gamma^2 F_G = \mathrm{Var}_\theta(q) \geq 0$, so $\log LO$ is non-decreasing and convex in γ, and it is strictly increasing in γ unless every edge of G joins vertices of equal degree. Moreover $\mathrm{Var}_\theta(q) = 0$ at some (equivalently, every) θ if and only if q_{uv} is constant over the edges of G, which covers regular graphs ($q \equiv 0$) and, for example, (Δ,δ)-semiregular bipartite graphs with $\Delta > \delta$. A bipartite graph is (Δ,δ)-semiregular when every vertex on one side has degree Δ and every vertex on the other side has degree δ, so every edge has degree pair (δ,Δ). On these every edge carries the same nonzero imbalance, and $\log LO$ is affine and strictly increasing in γ. F_G is strictly convex if and only if the edge feature vectors $\{z_{uv}\}_{uv\in E}$ do not lie in a proper affine subspace of $\mathbb{R}^3$. On a regular graph $F_G(\theta) = \log m + \theta^\top z_0$ is affine and $\mathrm{Cov}_\theta(z) = 0$.

The proof is the standard computation of the gradient and Hessian of a log-sum-exp function and is given in Supplementary Section S10.

3.1 The ratio plane, crossings and identifiability

Since $4ij/(i+j)^2 = 1 - q_{ij}^2$, every admissible pair satisfies $\ln(ij) = 2\ln(i+j) + \ln(1-q_{ij}^2) - \ln 4$, and therefore

$$LO(G;\alpha,\beta,\gamma) = 4^{-\alpha} \sum_{uv\in E(G)} (d_u + d_v)^{2\alpha+\beta}\,(1-q_{uv}^2)^{\alpha}\, e^{\gamma q_{uv}}. \qquad (4)$$

With $\kappa = 2\alpha + \beta$, the parameter κ acts only on the degree sum $d_u + d_v$, and α, γ act only on q_{uv}, a function of the degree ratio d_u/d_v.

Definition 3 (Ratio plane)**.** The ratio plane is $\Pi = \{(\alpha,\beta,\gamma) \in \mathbb{R}^3 : \beta = -2\alpha\}$, the plane $\kappa = 0$. It contains the pure-γ points $(0,0,\gamma)$ and the points $(\frac{1}{2},-1,0)$ and $(-\frac{1}{2},1,0)$, at which $GA = 2\,LO(\frac{1}{2},-1,0)$ and $AG = \frac{1}{2}LO(-\frac{1}{2},1,0)$.

Graphs with equal profiles receive equal values of every vertex-degree-based index [37,38], and sufficient linear criteria for an index to separate all distinct profiles within a class of graphs of fixed order are given in [38,39]. The theorem below concerns a coarser loss. On Π distinct profiles with equal q-histograms receive equal values. Part (iii) shows that such profiles occur among trees of every order from 13 (from 16 when $\Delta \le 4$); that no smaller tree has them is a computer-assisted result (see the remark on the floor on Π later in this subsection).

Theorem 2 (Ratio plane)**.** *Let G and H be graphs.*

(i) On Π, $LO(G;\alpha,-2\alpha,\gamma) = \sum_q n_q(G)\left((1-q^2)/4\right)^{\alpha} e^{\gamma q}$ depends on G only through its q-histogram, and G and H agree at every point of Π if and only if $n_q(G) = n_q(H)$ for all q. Otherwise their values differ at every point of Π outside a set of measure zero.

(ii) For every nonzero algebraic γ, $LO(G;0,0,\gamma) = LO(H;0,0,\gamma)$ if and only if $n_q(G) = n_q(H)$ for all q. On any finite set of graphs the same holds for all but finitely many real γ.

(iii) Let G and H be trees with distinct edge-degree-pair profiles and equal q-histograms, each having a leaf adjacent to a vertex of the same degree $d \geq 2$. Attaching a pendant path of length $L \geq 1$ at such a leaf in each, so that the leaf becomes a vertex of degree 2, produces trees with the same two properties and the same maximum degree. The profiles change by the same vector, whether or not the chosen leaf edge was the only $(1, d)$-edge. Consequently, for every $n \geq 13$ (and every $n \geq 16$ among trees with $\Delta \leq 4$) there are non-isomorphic trees of order n with distinct degree-pair profiles that no point of Π separates.

Proof. (i) The formula is immediate from (4), since setting $\beta = -2\alpha$ removes the degree-sum factor, and it shows that equal histograms give agreement at every point of Π. For the measure-zero clause, suppose the histograms differ. At $\alpha = 0$ the difference is $\sum_q (n_q(G) - n_q(H))\, e^{\gamma q}$, an exponential sum with distinct exponents and not all coefficients zero, so it is not identically zero. This uses the first of two standard facts: a nonzero exponential sum with distinct real exponents is not identically zero, because as $\gamma \to +\infty$ the term with the largest exponent dominates. The difference on Π is therefore a nonzero real-analytic function of (α, γ), whose zero set has measure zero. This is the second fact: the zero set of a nonzero real-analytic function on $\mathbb{R}^2$ has Lebesgue measure zero [30]. (ii) Agreement for equal histograms is immediate from the formula in (i). Since degrees are integers, every imbalance q is rational, so for algebraic $\gamma \neq 0$ the exponents γq, $q \in \mathbb{Q}$, are distinct algebraic numbers. The Lindemann–Weierstrass theorem [5] says that for distinct algebraic numbers $x_1, \ldots, x_k$ the exponentials $e^{x_1}, \ldots, e^{x_k}$ are linearly independent over the field of algebraic numbers. Hence the numbers $e^{\gamma q}$ are linearly independent over the algebraic numbers, and the integer coefficients $n_q(G) - n_q(H)$ must all vanish. For real γ, a nonzero exponential sum with k distinct real exponents has at most $k - 1$ real zeros (Proposition 4 below), and a finite set of graphs gives finitely many such sums. (iii) The last clause follows from (i) once the trees exist, since equal histograms give agreement on all of Π. The attachment, a device also used by Rada [38] for edge-type vectors, changes the profile of each tree by the same vector,

$-(1,d)+(2,d)+(L-1)(2,2)+(1,2)$, so distinct profiles stay distinct and equal histograms stay equal. The new or changed vertices have degree at most 2, so the maximum degree is unchanged when it is at least 2. Trees realising the two order-13 profiles listed in Section 4.2 are the following. Their edge lists confirm the common q-histogram displayed there: one edge of imbalance 0, two of $\frac{1}{5}$, one of $\frac{1}{3}$, three of $\frac{1}{2}$ and five of $\frac{5}{7}$. T_1: a path $c\,a\,b\,d\,e\,\ell$ in which c carries five pendant leaves, b and d carry one pendant leaf each and ℓ is a leaf, so that $d_c = 6$, $d_a = d_e = 2$, $d_b = d_d = 3$; its edges are $(1,6)^5, (2,6), (2,3)^2, (3,3), (1,3)^2, (1,2)$. T_2: a path $c\,b\,x\,y\,z$ in which c carries five pendant leaves, b one and z two, so that $d_c = 6$, $d_b = d_z = 3$, $d_x = d_y = 2$; its edges are $(1,6)^5, (3,6), (1,3)^3, (2,3)^2, (2,2)$. Both have a leaf adjacent to a vertex of degree $d = 3$. Consider now the order-16 pair with $\Delta \le 4$. The profile $\{(1,2)^2, (1,4)^8, (2,4)^2, (4,4)^3\}$ is realised by a path $A\,B\,C\,D$ of vertices of degree 4. Here A and D carry three pendant leaves each, while B and C carry one pendant leaf and one pendant path of length 2 each. The profile $\{(1,4)^8, (2,2)^3, (2,4)^4\}$ is realised by three vertices A, B, C of degree 4. They are joined as $A\,w\,B$ through a vertex w of degree 2 and as $B\,v_1v_2v_3v_4\,C$ through a path of four vertices of degree 2. The vertices A, B and C carry three, two and three pendant leaves. Both trees have the q-histogram $\{0^3, \frac{1}{3}^4, \frac{3}{5}^8\}$, since the pairs $(1,2)$ and $(2,4)$ have $q = \frac{1}{3}$, the pair $(1,4)$ has $q = \frac{3}{5}$ and the pairs $(2,2)$ and $(4,4)$ have $q = 0$. Both also have a leaf adjacent to a vertex of degree $d = 4$, which gives every larger order. ■

For graphs of equal order, part (ii) is Theorem 3.4 of [37] applied to the algebraic edge weights γq_{ij}, whose classes of equal weight are the q-classes: two graphs receive the same value for every algebraic $\gamma \neq 0$ exactly when they have equal q-histograms. The proof above does not need the order restriction, as the pair K_4, $K_{1,3}$ in the remark after Proposition 3 illustrates. The propagation by a pendant path in part (iii) is the device of [38]. The construction parallels Example 2.2 of [39], where indices that depend only on the degree sum are defeated by chemical trees whose counts agree on classes of equal sum. Here the classes are those of equal ratio, which for $\Delta \le 4$ are $\{(1,2), (2,4)\}$ and $\{(2,2), (3,3), (4,4)\}$ (ignoring $(1,1)$, which occurs only in K_2).

Song and Huang [48] studied a class-wide question of the same kind for the edge division vector of a tree. It counts the edges e of a tree T by the order $\min\{n_u(e), n_v(e)\}$ of the smaller component of $T - e$ and so controls every edge-additive eccentric index, such as the Wiener index. They determine all such trees of order at most 10. Neither partition of trees refines the other: the two seven-vertex trees with edge division vector $(4, 1, 1)$ have different degree-pair profiles and different q-histograms, while the two octane pairs of Table 1 have different edge division vectors and different Wiener indices. Explicit trees with equal Randić index and different class counts were given by Rada [37, Example 2.6], on 29 vertices, and as an infinite family of chemical trees by Rada, Rodríguez and Sigarreta [39, Example 2.1].

We found no earlier determination of the orders 13 and 16, or of explicit colliding trees for degree-ratio indices, in a literature search recorded in the repository [33]. The minimality of orders 13 and 16 is computer-assisted. It rests on an exhaustive enumeration of all trees of orders 2 to 17 in exact rational arithmetic, reproducible with the code of the repository [33].

Corollary (Degree-ratio indices)**.** *Let $\rho_{uv} = \min(d_u, d_v)/\max(d_u, d_v)$ and let $I_f(G) = \sum_{uv \in E(G)} f(\rho_{uv})$ for any function f on $(0, 1]$. Since $\rho_{uv} = (1 - q_{uv})/(1 + q_{uv})$, I_f depends on G only through its q-histogram. Consequently the trees of Theorem 2(iii), which exist at every order $n \geq 13$ (and $n \geq 16$ with $\Delta \leq 4$), are not separated by any such index. This applies to GA, AG, the exponential geometric–arithmetic index [11], the diminished Sombor index [32], the symmetric division deg index [50], the inverse symmetric division deg index and its exponential [12], and the degree-ratio indices [19, 35], including the degree-ratio Sombor index studied in [4, 21, 45].*

Remark. The corollary follows from the identity for ρ_{uv}; it is the structure observation of [37, Proposition 2.5] for edge weights that depend only on the degree ratio, and what is new are the explicit witnesses and the orders at which they first occur. Within the class it describes, the pure-γ points at algebraic $\gamma \neq 0$ attain the most possible: by part (ii) they separate every pair of graphs with distinct q-histograms. GA does not (see the remark on the floor on Π below). The parent comes close: at $\gamma = 0$ part (i) gives

$\sum_q n_q\,((1-q^2)/4)^\alpha$, an exponential sum in α with distinct bases, which separates any two given distinct histograms at all but finitely many α, hence all pairs at once for α outside a countable set. It fails at $\alpha = \frac{1}{2}$, the point of GA. The limit thus belongs to the dependence on the degree ratio, not to γ, which adds a form valid at every algebraic value. The orders 13 and 16 are the first at which the q-histogram, and hence every degree-ratio index simultaneously, fails to separate distinct profiles. A single index can fail earlier through coincidences among its values. The symmetric division deg index merges two distinct profiles of chemical trees already at order 9, for instance $\{(1,2),(1,3)^2,(2,2)^4,(2,3)\}$ and $\{(1,2)^3,(1,3),(2,3)^3,(3,3)\}$, both with value 58/3, and the inverse symmetric division deg index does so at order 10 (exact enumeration in the repository).

Off Π the argument behind [37, Theorem 3.4] applies unchanged when the coefficients are algebraic rather than integer. The following proposition records this direct consequence of the Lindemann–Weierstrass theorem, which is used in Section 4.2; we do not claim it as a new theorem.

Proposition 3 (Parent distinctions survive)**.** *Let $\alpha,\beta \in \mathbb{Q}$ and let $\gamma \neq 0$ be algebraic. For graphs G, H and each imbalance value q write $c_q(G,H) = \sum_{(i,j)\colon q_{ij}=q}\big(m_{ij}(G) - m_{ij}(H)\big)(ij)^\alpha(i+j)^\beta$. Then $LO(G;\alpha,\beta,\gamma) = LO(H;\alpha,\beta,\gamma)$ if and only if $c_q(G,H) = 0$ for every q. In particular, if $M_{\alpha,\beta}(G) \neq M_{\alpha,\beta}(H)$ then $LO(G;\alpha,\beta,\gamma) \neq LO(H;\alpha,\beta,\gamma)$: at rational exponents a nonzero algebraic γ never destroys a distinction made by the parent.*

Proof. The difference is $\sum_q c_q(G,H)\,e^{\gamma q}$, a sum over the distinct imbalances q occurring in G or H. For rational α,β the numbers $(ij)^\alpha(i+j)^\beta$ are algebraic, being rational powers of positive integers, so the coefficients $c_q(G,H)$, integer combinations of them, are algebraic. For algebraic $\gamma \neq 0$ the exponents γq are distinct algebraic numbers, so by the Lindemann–Weierstrass theorem [5] the sum vanishes only if every $c_q(G,H)$ vanishes. The last claim follows because $\sum_q c_q(G,H) = M_{\alpha,\beta}(G) - M_{\alpha,\beta}(H)$. ∎

Remark. At rational α, β, two graphs merged by $M_{\alpha,\beta}$ stay merged at an algebraic $\gamma \neq 0$ only if every class sum $c_q(G,H)$ vanishes, which is stronger than $M_{\alpha,\beta}(G) = M_{\alpha,\beta}(H)$. On the trees of order 16, $\gamma = 1$ recovers the

distinctions the parent loses at $(0, \frac{1}{2}, 0)$ and $(0, \frac{1}{3}, 0)$ (Section 4.2). The algebraic hypothesis cannot be dropped: at $\gamma = \ln 4$, $LO(K_4; 0, 0, \gamma) = 6 = 3e^{\gamma/2} = LO(K_{1,3}; 0, 0, \gamma)$.

Remark (The floor on Π). By exhaustive enumeration of all trees of orders 2–17 in exact arithmetic (Table 2), two trees of order at most 12 with equal q-histograms have equal degree-pair profiles, and the same holds for trees with $\Delta \leq 4$ of order at most 15. With part (iii), the pure-γ points at algebraic $\gamma \neq 0$ attain the degree-pair floor of Section 4.1, separation of all distinct profiles, exactly up to those orders and at no larger order. Among all trees, at orders 13–17, generic points off Π (Proposition 6) and the Randić index take more distinct values on these trees than any point of Π. A point of Π can also do worse. GA merges two distinct q-histograms among trees first at order 17, for instance $\{(1,5)^{10}, (2,2)^2, (5,5)^2, (2,5), (1,2)\}$ and $\{(4,5)^3, (1,4)^5, (1,5)^6, (2,5), (1,2)\}$, which have equal GA and different q-histograms.

Proposition 4 (Order exchanges along γ)**.** *Fix (α, β) and graphs G, H, let $q_1 < \cdots < q_r$ be the distinct imbalances of the degree pairs occurring in G or H, and write $D(\gamma) = LO(G; \alpha, \beta, \gamma) - LO(H; \alpha, \beta, \gamma) = \sum_{k=1}^{r} c_k e^{\gamma q_k}$ with $c_k = \sum_{q_{ij}=q_k} (m_{ij}(G) - m_{ij}(H))(ij)^\alpha (i+j)^\beta$. Let V be the number of sign changes in $(c_1, \ldots, c_r)$ after zero terms are removed. If $D \not\equiv 0$, then D has at most $V \leq r - 1$ real zeros counted with multiplicity, and their number has the parity of V. In particular, for graphs with $\Delta \leq 4$ the possible imbalances are $0, \frac{1}{7}, \frac{1}{5}, \frac{1}{3}, \frac{1}{2}, \frac{3}{5}$, and two such graphs exchange order at most five times as γ ranges over $\mathbb{R}$, for every (α, β).*

Proof. With $x = e^\gamma > 0$, D becomes the generalised polynomial $\sum_k c_k x^{q_k}$ with distinct real exponents, so Laguerre's extension of Descartes' rule of signs applies [25]. Consider a finite sum $\sum_k c_k x^{q_k}$ with real exponents $q_1 < \cdots < q_r$ and let V count the sign changes in its coefficient sequence. The rule says that the number of zeros in $x > 0$, counted with multiplicity, is at most V and differs from V by an even number. Zeros of D in γ correspond one-to-one, with multiplicity, to positive zeros in x. The number of real zeros of D, with multiplicity, is therefore at most V and has its parity. As an independent check, the parity also follows from the signs of D as

$\gamma \to \mp\infty$, which are the signs of the first and last nonzero c_k. Divide D by $e^{\gamma q_k}$ for the first k with $c_k \neq 0$ and let $\gamma \to -\infty$, or for the last such k and let $\gamma \to +\infty$. Only c_k survives. ■

Remark (Sharpness). The bound V is attained. The trees of order 12 with profiles $\{(1,3)^2, (1,4)^2, (1,6)^5, (3,4), (4,6)\}$ and $\{(1,2), (1,5)^4, (1,6)^4, (2,6), (5,6)\}$ have $\Delta = 6$, eight distinct imbalances and $V = 5$, and at $(\alpha, \beta) = (-1, 0)$ they cross at $\gamma \approx -17.2$, -12.0, -2.0, 2.7 and 31.1, which are therefore all zeros of D. Here $V = 5 < r - 1 = 7$. The first tree is a vertex of degree 6 with five leaves and a neighbour of degree 4, which carries two leaves and a vertex of degree 3 with two leaves. The second is a vertex of degree 6 with four leaves, a neighbour of degree 5 with four leaves, and a neighbour of degree 2 ending in a leaf. Whether five crossings are attained for $\Delta \leq 4$ is open. Along a line $\theta + tv$ not parallel to the γ axis the exponents of D are the values $v^\top z_{ij}$, of which there can be as many as admissible pairs, so the constant five is specific to the γ axis.

Per-graph identifiability, the directions along which the index of a single graph merely rescales, and the limits of LO along the γ axis are treated in Supplementary Section S10.

Remark (Data sets). For a QSPR data set the relevant set is the union of the pairs realised by its molecules. A direction that rescales every molecule's index by the same factor leaves every correlation unchanged and is absorbed into a regression slope. The affine dimension of a set of vectors z_{ij} is the dimension of the smallest affine subspace of $\mathbb{R}^3$ containing them. On the octanes this union contains nine of the ten admissible pairs and its affine dimension is 3, so the octane fits of Section 6 are free of exact confounding of this kind.

4 Discrimination

4.1 The degree-pair ceiling

No vertex-degree-based index separates graphs with equal degree-pair counts, so the number of distinct profiles is a *ceiling* on distinct values and a *floor* on degeneracy.

Proposition 5 (Degree-pair ceiling)**.** *If two graphs G and H have identical unordered edge-degree-pair count vectors, $m_{ij}(G) = m_{ij}(H)$ for all $i \le j$, then $LO(G;\theta) = LO(H;\theta)$ for every $\theta \in \mathbb{R}^3$, and more generally $TI(G) = TI(H)$ for every vertex-degree-based index TI. Among the* 18 *octane isomers exactly* 16 *distinct count vectors occur, and the pairs {3-methylheptane, 4-methylheptane} and {3,4-dimethylhexane, 3-ethyl-2-methylpentane} share their count vectors but differ in measured properties (Table 1). Hence every vertex-degree-based descriptor, tuned or not, carries a nonzero irreducible in-sample error on this benchmark.*

Proof. By Definition 1, $LO(G;\theta)$ depends on G only through the multiset of unordered edge degree pairs, that is, through $(m_{ij}(G))_{i\le j}$, and the same holds for every vertex-degree-based index [38], since these graphs share the M-polynomial [13]. The counts for the 18 octane isomers are computed and cross-checked in the reproducibility package, and Table 1 lists the common count vectors of both collision pairs. ■

A function of the count vector predicts one value per collision class, so its sum of squared errors is at least the within-class sum of squares. For the octane boiling points this floor is $2.925\,^{\circ}\mathrm{C}^2$, a root mean square of $0.403\,^{\circ}\mathrm{C}$. It bounds a single fixed predictor, not the cross-validated errors of Section 6, where each fold fits its own.

Remark (Dimension of the counts). For a tree of order $n \ge 3$ with $\Delta \le 4$ the ten counts are not free. A $(1,1)$ edge would be a component K_2, so $m_{11} = 0$. There are $n-1$ edges, and since a vertex of degree d lies on d edges, $\sum_{i\le j}(1/i+1/j)\,m_{ij} = n$ (see [10, eqs. (3)–(4)]). The nine remaining counts therefore have affine dimension at most 7 at fixed n and at most 8 when n varies. Both bounds are attained by the data sets of Section 6: the centred count matrices of the 18 octanes and the 34 measured decanes have rank 7, and that of the 100 pooled alkanes has rank 8.

The ceiling is tight. Distinct count vectors are separated at almost every parameter choice.

Proposition 6 (Generic discrimination)**.** *Fix $\Delta \ge 1$ and let $m \ne m'$ be two distinct edge-degree-pair count vectors supported on the admissible*

pairs P_Δ. Then $D(\theta) = \sum_{(i,j)\in P_\Delta}(m_{ij} - m'_{ij})\, e^{\theta^\top z_{ij}}$ is a real-analytic function of θ that is not identically zero, and its zero set has Lebesgue measure zero in $\mathbb{R}^3$ [30]. Consequently, any two graphs with distinct count vectors are distinguished by $LO(\cdot\,;\theta)$ for almost every $\theta \in \mathbb{R}^3$, while only graphs with identical count vectors (Proposition 5) are indistinguishable for all θ.

Proof. The map $(i,j) \mapsto z_{ij}$ is injective on P_Δ, since $\ln(ij)$ and $\ln(i+j)$ recover the product and the sum, which determine the unordered pair. Choose $v \in \mathbb{R}^3$ such that the inner products $v^\top z_{ij}$ are pairwise distinct, which excludes only finitely many hyperplanes of v. Along the ray $\theta = tv$, $D(tv) = \sum_k c_k e^{t\lambda_k}$ with distinct exponents λ_k and coefficients $c_k = m_{i_k j_k} - m'_{i_k j_k}$ that are not all zero, and such an exponential sum vanishes identically only if all $c_k = 0$. Hence $D \not\equiv 0$ on the ray, and so on $\mathbb{R}^3$. Since D is a finite sum of exponentials of linear functionals, it is real-analytic on $\mathbb{R}^3$, and the zero set of a nonzero real-analytic function has Lebesgue measure zero. ∎

In the terminology of Rada, Rodríguez and Sigarreta [39], an index is weakly discriminating over a class when equal values force equal count vectors. They showed that weights linearly independent over $\mathbb{Z}$, for instance exponentials of distinct algebraic numbers, give weakly discriminating indices [39, Theorems 3.1 and 3.8], and support restrictions of this criterion are given in [38, Theorem 3, Corollary 2]. As there are countably many pairs of count vectors, Proposition 6 gives that almost every point of the family is weakly discriminating over every class at once. The proof uses only the injectivity of $(i,j) \mapsto (\ln(ij), \ln(i+j))$, so Proposition 6 is inherited from the parent $M_{\alpha,\beta}$ rather than conferred by γ. The parameter γ adds no graph information beyond the counts, of which every vertex-degree-based index is a functional [38], only the different weighting of Lemma 1.

At the rational exponents used in practice the parent can fail: the sum-connectivity index $\chi = M_{0,-1/2}$ and the index $M_{0,1/2}$ are not weakly discriminating over chemical trees [39, Example 2.2]. The imbalance coordinate separates the pair of trees used there for χ.

Example 1. For $t \geq 1$ the chemical trees S and T of order $17 + t$ in Example 2.2 of [39] have $m(S)-m(T) = 6(1,3)+4(2,3)+4(3,3)-4(1,4)-6(2,2)-4(2,4)$ and equal sum-connectivity index. At $(\alpha, \beta) = (0, -\frac{1}{2})$,

$$D(\gamma) = LO(S) - LO(T) = 3\big(e^{\gamma/2} - 1\big) - \tfrac{4}{\sqrt{5}}\big(e^{3\gamma/5} - e^{\gamma/5}\big) + \tfrac{4}{\sqrt{6}}\big(1 - e^{\gamma/3}\big),$$

so $D(0) = 0$ and $D(1) \approx 0.2255$. The separation is small for small γ, for instance $D(10^{-6}) \approx 2.4 \times 10^{-7}$.

Table 1. The two degree-pair collision pairs among the 18 octane isomers. Each pair shares its complete edge-degree-pair count vector, shown as multiplicities of the unordered pairs, and is therefore indistinguishable by every vertex-degree-based index for all parameter values. The two members differ in the measured properties. Units: T_B in °C, ΔH_f and ΔH_{vap} in kcal mol^{-1}, S in cal mol^{-1} K^{-1}; ω is dimensionless.

Isomer	T_B	ΔH_f	ΔH_{vap}	S	ω
shared count vector $2(1,2)$, $1(1,3)$, $2(2,2)$, $2(2,3)$:					
3-methylheptane	118.9	−50.82	9.52	111.26	0.371
4-methylheptane	117.7	−50.69	9.48	109.32	0.372
shared count vector $2(1,2)$, $2(1,3)$, $2(2,3)$, $1(3,3)$:					
3,4-dimethylhexane	117.7	−50.91	9.32	106.59	0.340
3-ethyl-2-methylpentane	115.6	−50.48	9.21	106.06	0.330

4.2 Degeneracy across tree orders

Degeneracy [26] measures the failure to discriminate non-isomorphic graphs, and we quantify it with the distinct-value convention $100(1-k/N)$, where k is the number of distinct values among $N = 106$ trees (Supplementary Section S5). It is not the share of trees whose value another tree also takes, which is larger: for $LO(0,0,1)$, 45 of the 106 trees (42.5%). On the 106 trees of order 10 the pure-γ points $LO(0,0,1)$ and $LO(0,0,2)$ attain 25.5% (79 distinct values), tied with R, GA and AG. The 106 trees realise only 79 distinct edge-degree-pair count vectors, so 25.5% is the floor for *any* degree-pair-based index, while M_1 (83.0%) and M_2 (67.0%) are the most degenerate (Figure 1). For orders 7 through 12, on all trees and on those with $\Delta \leq 4$ (Supplementary Table S4), the pure-γ points

and R attain the floor at every order. The index χ leaves it from order 10, with 77 values against 79 profiles. Theorem 2 makes this exact. The pure-γ points separate precisely the q-histograms, which coincide with the degree-pair profiles up to order 12 (order 15 for $\Delta \leq 4$) and not beyond. At order 13 the profiles $\{(1,2)^1,(1,3)^2,(1,6)^5,(2,3)^2,(2,6)^1,(3,3)^1\}$ and $\{(1,3)^3,(1,6)^5,(2,2)^1,(2,3)^2,(3,6)^1\}$ (exponents are edge multiplicities) share the q-histogram $\{0^1,\frac{1}{5}^2,\frac{1}{3}^1,\frac{1}{2}^3,\frac{5}{7}^5\}$. Table 2 collects the first failures, and the index R still attains the floor at order 13 (570 values) and leaves it at order 14, where the pure-γ points give 1090 values.

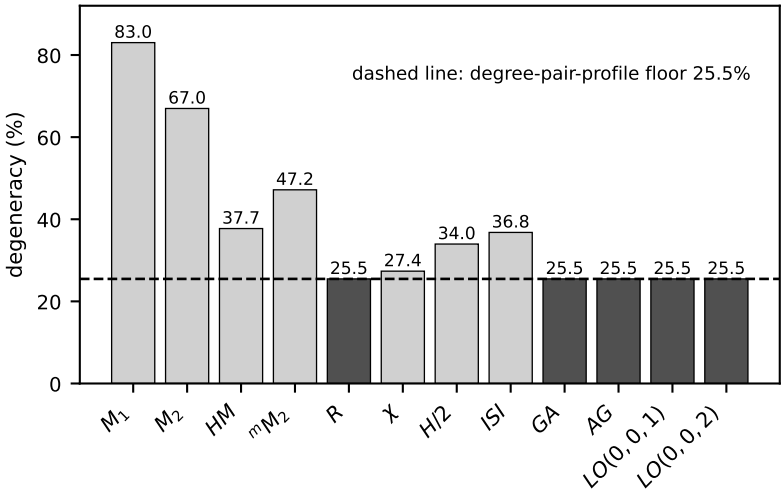


Figure 1. Degeneracy percentage of the compared indices on the 106 non-isomorphic trees of order 10. The dashed line marks the degree-pair-profile floor, $25.5\% = 100(1 - 79/106)$. The 106 trees realise only 79 distinct edge-degree-pair count vectors, so no vertex-degree-based index can fall below it; darker bars mark the indices on the floor. An index on the floor separates every pair of trees with distinct profiles at this order; it carries no information beyond the profile, and the pure-γ points share the floor with R, GA and AG.

Off the ratio plane the imbalance factor can restore discrimination lost at particular classical exponents. A point $(0,\beta,0)$ weights an edge by its degree sum only, so pairs with equal sum receive equal weight. With $\gamma \neq 0$ their imbalances separate them. On all trees of order 16 (4069 profiles),

Table 2. First tree order at which discrimination is lost. Values are counted as distinct at 11 significant digits. Recounts at 50 to 60 digits agree. Every count, ties included, is also certified exactly. For the pure-γ points this follows from the q-histograms (Theorem 2(ii)). For R and GA, including the merge of GA at order 17, it follows from exact arithmetic in the radicals $(ij)^a(i+j)^b$. This rests on the linear independence over $\mathbb{Q}$ of the k-th roots of distinct k-th-power-free positive integers [7]. "Below floor" is the first order with fewer distinct values than distinct edge-degree-pair profiles, the floor for every vertex-degree-based index. The entry gives (values/profiles) at that order. "Merges" is the first order at which an index on Π gives equal values to two trees with different q-histograms. This does not apply to R, which is not on Π. $LO(0,0,\gamma)$ denotes the pure-γ points $\gamma = 1, 2$. Orders 2 to 17 were examined. Equal q-histograms of distinct profiles were sought at every order from 2 and first found at the orders shown.

	all trees		trees with $\Delta \leq 4$	
Index	below floor	merges	below floor	merges
$LO(0,0,\gamma)$	13 (566/570)	never[a]	16 (809/810)	never[a]
R	14 (1095/1100)	n/a[b]	17 (1191/1194)	n/a[b]
GA	13 (566/570)[c]	17[d]	16 (809/810)[c]	> 17[d]

[a] All q-histograms are separated for every nonzero algebraic γ (Theorem 2(ii)). [b] R is not on Π. [c] GA lies on Π and merges no q-histograms below order 17, so there it ties the pure-γ points. [d] The witness (Section 3.1) has maximum degree 5; among trees with $\Delta \leq 4$, GA merges no q-histograms up to order 17, the largest examined.

$LO(0, \frac{1}{2}, 0)$ and $LO(0, \frac{1}{3}, 0)$ take only 3475 and 3476 distinct values, while $LO(0, \frac{1}{2}, 1)$ and $LO(0, \frac{1}{3}, 1)$ separate all 4069 profiles. $LO(\frac{1}{2}, 0, \cdot)$ rises from 3940 to 4067 and the Randić point $LO(-\frac{1}{2}, 0, \cdot)$ from 4007 to 4068 as γ goes from 0 to 1. Points on Π such as GA are unaffected (3996 in both cases), consistent with Theorem 2, and generic parent points such as $LO(0.3, 0.2, 0)$ already separate all 4069 profiles (Proposition 6). These counts are exact for orders 12–16 (at $\gamma = 1$ by Proposition 3 and [7]), but the effect is not proved for larger orders. It yields no predictive gain over a tuned parent in Section 6, and at fixed fractional exponents the sign of the change depends on the property (Supplementary Section S5).

4.3 Structure sensitivity

The structure-sensitivity criterion of Furtula, Gutman and Dehmer [14], with a fingerprint-based variant due to Rakić and Furtula [40], was implemented with the published protocol on the 75 decane isomer trees. For a tree, the similar structures are those obtained by relocating a single edge within the molecular class. SS and Abr are the class means of the mean and the maximum relative change of the index over these neighbours. The implementation reproduces the published decane-class values of Barman and Das [6], with eleven of fourteen SS/Abr values agreeing exactly at the four displayed decimals. Each of the other three agrees within one unit in the fourth. The protocol and this control are detailed in Supplementary Section S6. Under the published desideratum (SS large, Abr small), $LO(0, 0, 2)$ has a structure sensitivity indistinguishable at three decimals from that of the hyper-Zagreb index (0.1257 against 0.1256). Its abruptness is markedly smaller (0.274 against 0.313). The sensitivity-to-abruptness ratios of the pure-γ points (0.4579 and 0.4565) are comparable to those of $H/2$, GA and R (0.448–0.451, Table 3). In view of the redundancy noted in Section 6.1, these third-decimal differences do not rank the indices. SS also grows with γ along the pure-γ line, and the parameter-free aggregate $IRLA$, which these points approximate to first order, has a higher ratio (Supplementary Section S6), so the equality of SS with hyper-Zagreb at $\gamma = 2$ reflects the chosen value of γ.

Table 3. Structure sensitivity SS, abruptness Abr, and their ratio on the 75 decane isomers under the published similar-structure protocol of [14] (single-edge relocation within the molecular class).

Index	SS	Abr	SS/Abr
M_1	0.0566	0.1336	0.4238
M_2	0.0922	0.2443	0.3776
HM	0.1256	0.3126	0.4017
${}^{m}M_2$	0.0516	0.1177	0.4384
R	0.0264	0.0588	0.4481
χ	0.0264	0.0591	0.4463
$H/2$	0.0516	0.1143	0.4514
ISI	0.0226	0.0597	0.3795
GA	0.0241	0.0535	0.4507
AG	0.0256	0.0579	0.4415
$LO(0,0,1)$	0.0624	0.1367	0.4565
$LO(0,0,2)$	0.1257	0.2745	0.4579

5 Extremal trees for the pure-imbalance points

Closed forms on standard graphs and termwise bounds, with their equality cases, are given in Supplementary Section S10.

Proposition 7 (Extremal trees of the pure-γ points)**.** *Let $n \geq 4$ and $\gamma > 0$. Among all trees of order n, the path P_n is the unique minimiser and the star S_n the unique maximiser of $LO(\cdot\,; 0, 0, \gamma)$, with*

$$(n-3) + 2e^{\gamma/3} \;\leq\; LO(T; 0, 0, \gamma) \;\leq\; (n-1)\, e^{\gamma(n-2)/n}.$$

For $\gamma < 0$ the roles of P_n and S_n are exchanged.

Proof. Let $\gamma > 0$. Every edge uv of a tree of order n has $q_{uv} \leq (n-2)/n$, with equality if and only if $\{d_u, d_v\} = \{1, n-1\}$, so $LO(T; 0, 0, \gamma) \leq (n-1)e^{\gamma(n-2)/n}$ with equality if and only if every edge is a $(1, n-1)$-edge, that is, $T = S_n$. For the lower bound, a tree with $k \geq 2$ leaves has at least k edges incident to a leaf, and each such edge has $q \geq \frac{1}{3}$ because the other end has degree at least 2 and $(d-1)/(d+1)$ is increasing in d. Hence $LO(T; 0, 0, \gamma) \geq (n-1-k) + ke^{\gamma/3} \geq (n-3) + 2e^{\gamma/3}$, since

$e^{\gamma/3} > 1$. Equality in the second inequality forces $k = 2$, so $T = P_n$, and P_n attains the bound because its two pendant edges are $(1,2)$-edges and its other edges are $(2,2)$-edges. For $\gamma < 0$ every inequality reverses. ■

The two bounds of Proposition 7 also follow from the criteria of Cruz and Rada [10]; the direct proof above is included because it is short and also gives the uniqueness of the extremal trees. For the weights $\varphi_{ij} = e^{\gamma q_{ij}}$ and $E = e^{\gamma/3}$, their function $f(i,j) = \frac{ij}{i+j}(\varphi_{ij}+2\varphi_{12}-3\varphi_{22})$ equals $2(E-1)$ at $(1,2)$ and $(2,2)$. For $\gamma > 0$ it is larger at every other pair: if $i,j \geq 2$ and $(i,j) \neq (2,2)$ then $\varphi_{ij}+2E-3 \geq 2(E-1) > 0$ and $ij/(i+j) > 1$, and for $(1,j)$, $j \geq 3$, one has $f \geq \frac{3}{4}(e^{\gamma/2}+2E-3) > 2(E-1)$, since with $x = e^{\gamma/6} > 1$ this reads $3x^3-2x^2-1 = (x-1)(3x^2+x+1) > 0$. Corollary 2.2 of [10] then gives the lower bound $LO(T;0,0,\gamma) \geq LO(P_n;0,0,\gamma)$, and their Corollary 3.2 gives the upper bound $LO(T;0,0,\gamma) \leq LO(S_n;0,0,\gamma)$, because $e^{\gamma q}$ is largest at the pair $(1,n-1)$. These corollaries state the bounds without equality cases. Uniqueness is given by the direct proof above, and also by the identities of their Theorems 2.1 and 3.1, since the inequalities just shown are strict away from the pairs $(1,2)$ and $(2,2)$, and $e^{\gamma q_{ij}} < e^{\gamma(n-2)/n}$ away from $(1,n-1)$. For $\gamma < 0$ every inequality reverses. Gao and Gao [15] give general sufficient conditions for the path to be the unique minimal tree and the star the unique maximal tree of an exponential vertex-degree-based index, a class to which $LO(\cdot\,;0,0,\gamma) = \sum_{uv} e^{\gamma q_{uv}}$ belongs. The star half also follows from the sufficient condition of [38, Section 4.2]; the path half does not follow from the corresponding condition there, because the $(1,2)$-weight $e^{\gamma/3}$ exceeds the $(2,3)$-weight $e^{\gamma/5}$. The proof uses only the bounds on q and does not extend verbatim to general (α,β), where the ordering of trees depends on the scale exponent $2\alpha+\beta$ as well as on γ. Among chemical trees ($\Delta \leq 4$) every edge has $q \leq \frac{3}{5}$, which gives $LO(T;0,0,\gamma) \leq (n-1)e^{3\gamma/5}$ for $\gamma > 0$, with equality only for $n = 5$. The path remains the unique minimiser for $\gamma > 0$ among chemical trees by Proposition 7, while the maximal chemical trees for $n > 5$ are left open. The orders 4 to 12 were also checked by enumeration in the reproducibility package.

6 QSPR diagnostics

This section asks whether the family, and γ in particular, describes or predicts alkane properties better than the classical indices it contains. The sets are small, so the octane correlations are descriptive, and the pre-registered tests on measured boiling points of Section 6.4 are the primary predictive evidence.

We compare the family with the classical indices it reduces to on the 18 octane isomers, following the benchmark of the diminished Sombor study [32], and use the 35 nonane isomers as a replication set. The comparison set consists of the exact reductions $M_1, M_2, HM, {}^mM_2, R, \chi, H/2, ISI, GA, AG$ and the fixed pure-γ points $LO(0,0,1)$ and $LO(0,0,2)$. The edge count $m = LO(0,0,0)$ is constant on the octanes and is omitted. No parameter in the correlation comparison is optimised against the property data, although the choice of $\gamma = 1, 2$ cannot be shown to precede the octane data (below).

Predictive performance is measured by the leave-one-out coefficient $Q^2 = 1 - \sum_i (y_i - \hat{y}_{(i)})^2 / \sum_i (y_i - \bar{y})^2$, where $\hat{y}_{(i)}$ predicts molecule i from a model fitted without it and $\bar{y}$ is the mean over all molecules. For a fixed descriptor, such as $LO(0,0,1)$ or $IRLA$, ordinary leave-one-out refits only the regression in each fold. Where a parameter point, an index or a ridge penalty is chosen from the data, nested leave-one-out repeats that choice in each fold on the training molecules alone, so the held-out molecule affects neither selection nor fit. An ordinary leave-one-out Q^2 of a point chosen on the same data omits the selection step and is therefore optimistic.

The repository history begins with the octane correlations already computed, so it cannot document that the benchmark points $\gamma = 1$ and 2 were fixed before the octane data were seen. An exploratory local-tuning check (Supplementary Section S2) exceeded the best fixed index only by 0.001 in sample on ΔH_{vap} and, in a leave-one-out correlation nested within globally chosen candidate pools, on octane T_B (0.764 against 0.745 for ${}^mM_2$). This is a small point-estimate difference without an interval, and no superiority is claimed.

6.1 Correlations on the octane isomers

Table 4 reports the Pearson correlation r between each index and the boiling point T_B, the enthalpy of formation ΔH_f, the enthalpy of vaporisation ΔH_{vap}, the entropy S and the acentric factor ω of the 18 octane isomers. The property values are the standard benchmark compilation, audited against the NIST WebBook [29] and the 1947 API Research Project 44 tables [44] (Supplementary Table S5 and Section S1).

The open items are seven entropies, the compiled acentric factors and the vaporisation enthalpy of 2,2,3,3-tetramethylbutane, which is crystalline at 298 K. The benchmark value 8.41 kcal mol^{-1} has no stated source and lies just below the estimate for a hypothetical supercooled liquid from sublimation minus fusion enthalpy (8.56). The NIST value of 10.26 at 298 K describes sublimation (Supplementary Section S8). In the ΔH_{vap} column $LO(0,0,1)$ stays first with the isomer excluded ($|r| = 0.980$), level at three decimals with $LO(0,0,2)$, and with the hypothetical-liquid estimate ($|r| = 0.976$), where $LO(0,0,2)$ follows at 0.970, while the sublimation enthalpy lowers every $|r|$ to 0.374 or below. These correlations are therefore conditional on a liquid-phase value for this isomer. The entropies can move the 0.001 margin between HM and M_1, so the ordering of the S column is not interpreted. None of the open items enters the tests of Sections 6.3 and 6.4.

The best correlation is split. ${}^{m}M_2 = LO(-1,0,0)$ is strongest on T_B and ΔH_f, $HM = LO(0,2,0)$ on S, $M_2 = LO(1,0,0)$ on ω and the fixed point $LO(0,0,1)$ on ΔH_{vap} ($|r| = 0.984$, with $LO(0,0,2)$ second at 0.981). At the exact-reduction triples the Loyola correlations coincide with the classical indices, as they must. Deleting single isomers shifts each bolded $|r|$ by at most 0.027 (Supplementary Section S1). The descriptors are nearly collinear. Fourteen pairs have $|r| \geq 0.99$, the first two principal components carry 99.05% of the variance (Supplementary Section S1), and the largest $|r|$ with a reduction is 0.982 for $LO(0,0,1)$ and 0.993 for $LO(0,0,2)$. Differences in the third decimal are therefore not rankings, and the bolded entries, each the largest of twelve correlations on 18 molecules, are optimistic.

Proposition 5 is realised on these data. The 18 isomers have only 16

Table 4. Pearson correlation r between the Loyola reductions, the fixed pure-γ benchmark points and the five physicochemical properties of the 18 octane isomers. Bold flags the largest $|r|$ per column, and this is descriptive. Margins to the runner-up are 0.001–0.032, and their paired bootstrap intervals include zero (Supplementary Section S1). These intervals are post-selection and not adjusted for multiplicity. The entry -0.000 (ISI, ΔH_f) is -0.0002.

Index	T_B	ΔH_f	ΔH_{vap}	S	ω
$M_1 = LO(0,1,0)$	-0.718	-0.762	-0.935	-0.970	-0.970
$M_2 = LO(1,0,0)$	-0.497	-0.542	-0.811	-0.934	$\mathbf{-0.988}$
$HM = LO(0,2,0)$	-0.654	-0.710	-0.903	$\mathbf{-0.971}$	-0.982
${}^{m}M_2 = LO(-1,0,0)$	$\mathbf{+0.855}$	$\mathbf{+0.891}$	$+0.925$	$+0.863$	$+0.794$
$R = LO(-1/2,0,0)$	$+0.819$	$+0.850$	$+0.958$	$+0.932$	$+0.898$
$\chi = LO(0,-1/2,0)$	$+0.800$	$+0.832$	$+0.961$	$+0.947$	$+0.924$
$H/2 = LO(0,-1,0)$	$+0.821$	$+0.849$	$+0.961$	$+0.932$	$+0.900$
$ISI = LO(1,-1,0)$	-0.005	-0.000	-0.383	-0.593	-0.738
$GA = 2\,LO(1/2,-1,0)$	$+0.822$	$+0.858$	$+0.965$	$+0.939$	$+0.906$
$AG = \frac{1}{2}LO(-1/2,1,0)$	-0.816	-0.859	-0.957	-0.938	-0.900
$LO(0,0,1)$	-0.828	-0.832	$\mathbf{-0.984}$	-0.928	-0.929
$LO(0,0,2)$	-0.829	-0.846	-0.981	-0.937	-0.925

distinct count vectors, and the members of the two collision pairs of Table 1 differ by 1.2 °C and 2.1 °C in T_B although every vertex-degree-based index gives them equal values.

6.2 Effect of freeing γ and of the search box

We ask whether, at identical search budgets, freeing γ improves out-of-sample prediction over an optimised parent $M_{\alpha,\beta}$. In each outer fold of a nested leave-one-out (Supplementary Section S3), triples (α, β, γ) for the Loyola model (LO) and pairs (α, β) with $\gamma = 0$ for the parent (GM) are drawn uniformly from a search box at equal budget. The candidate with the smallest inner leave-one-out error is refit and predicts the held-out isomer.

In the box $(-2, 2)$ freeing γ improved the nested Q^2 for ΔH_{vap} in 100/100 candidate streams at budgets 200 and 500, with median $\Delta Q^2 = +0.069$ and $+0.068$ (Supplementary Table S3), and in 93/100 and 96/100 with 2,2,3,3-tetramethylbutane excluded. It usually degraded ΔH_f, S and

ω, and was mixed on T_B; on ΔH_f the molecule-bootstrap interval for LO minus GM excludes zero against γ (Supplementary Section S4).

For ΔH_{vap}, however, the parent chose $(\alpha, \beta) = (1.021, -1.993)$, on the boundary of the box, in 16 of the 18 folds. By $\ln(ij) = 2\ln(i+j) + \ln(1-q^2) - \ln 4$ (Eq. (4)) the parent reaches imbalance information through $(1-q^2)^{\alpha}$ along $\beta \approx -2\alpha$, which needs large exponents and is the direction the box truncates.

The box of half-width 12, chosen after inspecting the parent's nested Q^2 on a dense grid as the smallest examined in which the parent's selections are interior, is used below for every property (details in Supplementary Section S11).

The matched comparison was repeated, without pre-registration, with 2000 candidates per model, seeds 0–49 and boxes of half-width 2 and 12 (Supplementary Table S6). In the wide box the vaporisation-enthalpy gain shrinks to about zero. A hybrid control widens only (α, β), keeps $\gamma \in (-2, 2)$ and is drawn after the parent from the same candidate stream. It gives median $\Delta Q^2 = +0.004$ on the octanes (28 of 50 seeds) and -0.002 on the nonanes (24 of 50), while widening all three coordinates gives -0.010 and -0.016. Most of the change comes from the parent improving in the wide box ($0.887 \to 0.942$ on the octanes), the rest from the three-dimensional search being slightly less efficient there ($0.957 \to 0.946$ for the hybrid control). On the other properties no advantage reaches 40 of 50 seeds in either box, and octane S moves from 11 to 38 wins, which shows how strongly the win count depends on the search protocol.

Under these protocols tuning γ gives no demonstrated advantage. Equal budgets do not give equally thorough searches in two and three dimensions, so the design may cost the three-parameter model part of the effect sought.

6.3 Replication on the nonane isomers

Under a protocol committed to the reproducibility package (commit `964b127`) before the nonane data were committed and before any model was fitted to them, the nested design was applied to the 35 constitutional nonane isomers. Boiling points were 35 values from the NIST WebBook

(NIST averages where given, otherwise the most recent determination, a deviation recorded in the protocol). Standard vaporisation enthalpies were 34 values from the NIST WebBook [29], of which 29 are compiled values: NIST lists 4 under Labbauf et al. [27] (printed as Labauf) and 25 under Reid [42], a review of the handbook [52]. Agreement with them therefore measures consistency with compiled values. These compilations may rest partly on structural-increment estimates, which would favour a degree-pair-based descriptor (Supplementary Section S4).

The pre-registered verdict is that the γ effect *replicates* for ΔH_{vap}, with LO better than GM in 95/100 and 98/100 seeds and median $\Delta Q^2 = +0.048$ and $+0.053$, and is *mixed* for T_B (58/100 and 63/100). *Replicates* is the protocol's name for its seed-count rule, which the protocol itself describes as a summary of optimiser variance and not an inferential test. A post hoc molecule bootstrap conditional on the fitted predictions gives a 95% interval for LO minus GM on nonane ΔH_{vap} of $[-0.006, +0.113]$ at budget 200 (Supplementary Section S4), and this interval includes 0. Within-fold baselines are in Supplementary Table S7.

The protocol fixed the box of half-width 2, which binds the parent, and in the wide box the gain shrinks to about zero (median $\Delta Q^2 = -0.002$ under the hybrid control). We report the verdict as obtained. Its seed-count criterion counts optimiser variance on fixed data, which, used on its own, we regard as a design error. After this result we made the measured boiling points of Section 6.4 the primary predictive evidence, and their protocol keeps a seed count only together with a molecule-level interval.

Ridge regression on the degree-pair counts has higher point estimates than every index model on the nonanes, although its conditional interval against the Loyola model on ΔH_{vap} includes zero. The untuned point $LO(0, 0, 1)$ matches the tuned Loyola model on nonane ΔH_{vap} in point estimate, so tuning γ adds nothing there over $\gamma = 1$. These comparisons and that with the parameter-free $IRLA$ are in Supplementary Section S11.

6.4 Measured boiling points

Measured vaporisation enthalpies are too scarce for a confirmatory test of the vaporisation-enthalpy gain. Among the acyclic alkanes C4–C16,

NIST lists only 25 skeletons outside C_8 and C_9, among them one of the 75 decanes, with at least one calorimetric standard-state 298 K determination. Two tests therefore used measured boiling points, a property on which freeing γ had given mixed results, so they do not test the vaporisation-enthalpy gain directly. The values come from the NIST WebBook [29], extracted by a rule committed to the repository (commit `a5cf378`) before any model was fitted. The rule was written after the NIST pages of all acyclic C4–C16 alkanes had been downloaded to count the available values; no C6, C7 or C10 boiling point had entered any model. The rule admits all hexanes, heptanes and octanes, 34 of the 35 nonanes and 34 of the 75 decanes. The included and excluded decanes do not differ detectably in branching or imbalance (Supplementary Section S9), but with 34 and 41 molecules this does not show that the decane cohort is representative.

Test A refits and reselects inside each leave-one-out fold on the decanes. Test B pools all 100 molecules with the carbon number n_C as a covariate. The octane and nonane boiling points were used above, so Test B is a pooled size-adjusted analysis rather than a fresh external test. Both use the wide-box design of Section 6.2, with $(\alpha, \beta) \in (-12, 12)^2$ for the parent and the same box with $\gamma \in (-2, 2)$ for the Loyola model, 2000 candidates and seeds 0–49. The pre-registered criterion for a γ effect required at least 40 wins in 50 seeds and a molecule-bootstrap interval excluding zero (results in Table 5 and below), which we read as requiring the lower end of the two-sided 95% interval to exceed zero.

Both pre-registered verdicts are negative. On the decane cohort the median $\Delta Q^2 = -0.072$, LO is ahead in 15 of 50 seeds, and the 95% interval is $[-0.156, +0.001]$. In the pooled size-adjusted set the values are -0.0001, 25 of 50 and $[-0.003, +0.002]$. On the decanes the point estimate therefore favours the parent, and the Loyola model is behind in most seeds, but the interval includes zero, so the test establishes neither a gain nor a loss from γ.

In the original box of half-width 2, reported as the pre-registration requires, the two tuned models are level on the decanes (medians 0.469 and 0.465), and the Loyola model is marginally ahead on the pooled set (0.980 against 0.976). Seed-by-seed comparisons within and across the

Table 5. Measured boiling points (leave-one-out Q^2, nested wherever an index, a point or a penalty is selected). Test A: decane cohort, refitted within folds. Test B: pooled C6–C10 with the carbon number n_C in every index model (Class., $IRLA$, LO_1: n_C plus that descriptor); Ridge uses the counts alone, whose sum is $n_C - 1$. Base: training-fold mean (Test A) or n_C alone (Test B). Class.: best classical index chosen in each fold. LO_1: fixed $LO(0, 0, 1)$. Ridge: ridge regression on the degree-pair counts. GM, LO_h: medians over seeds 0–49 in the wide boxes, LO_h being the hybrid control with (α, β) in the wide box and $\gamma \in (-2, 2)$. ΔQ^2: median over seeds of the paired difference LO_h minus GM. The registered intervals are conditional on the fitted predictions and omit refitting and selection uncertainty (see below). Wins: seeds (of 50) in which LO_h beats GM.

Test	Base	Class.	$IRLA$	LO_1	Ridge	GM	LO_h	ΔQ^2	wins
decanes ($n = 34$)	−0.062	0.176	0.167	0.149	0.860	0.514	0.406	−0.072	15
C6–C10, with n_C ($n = 100$)	0.938	0.967	0.963	0.963	0.991	0.977	0.977	−0.0001	25

two boxes were not registered and carry no interval. On the pooled set the narrow-box Loyola model leads in most seeds, so these comparisons are inconclusive. The pooled cohort was corrected after an external audit, and the uncorrected cohort gives the same verdict. Both are reported in Supplementary Section S11.

An exploratory resampling reruns the whole nested procedure for both models on random subsamples of 80% of the molecules. Each replicate uses a single candidate stream, so the ranges also contain optimiser variance. On the decanes (500 subsamples) the median ΔQ^2 is -0.110 with central 95% range $[-0.510, +0.312]$. On the pooled set (200 subsamples) it is $+0.001$ with range $[-0.015, +0.008]$. Neither range supports a gain from γ. The two summaries differ in training size and seed averaging and are not directly comparable. Nor are the ranges confidence intervals for the full-sample ΔQ^2.

The selection-aware decane range is so wide that it neither supports nor excludes gains of practical size, the registered interval omits refitting and selection, and no smallest effect of interest or equivalence margin was registered. The registered contrast also gives LO_h three dimensions to search against two for GM with the same candidate budget, which may handicap the three-parameter model (Section 6.2). The negative verdicts

therefore mean that no gain was demonstrated under these protocols; they do not establish equivalence, and they do not show that γ never helps.

Ridge regression on the ten degree-pair counts reaches $Q^2 = 0.860$ on the decanes, where every single index stays between 0.149 and 0.514 (medians over seeds for the tuned models), and 0.991 on the pooled set. On the registered secondary question of Test B, $IRLA$ and $LO(0, 0, 1)$ do not beat size plus the best classical index (0.963 against 0.967), while both tuned models have higher point estimates (0.977; no interval was registered for this secondary question).

A permutation check on the decanes repeats the whole nested procedure, including the choice of the ridge penalty, for 200 random permutations of the boiling points. The permuted Q^2 has median -0.09, 95th percentile -0.01 and maximum 0.20. None reaches 0.860, so the empirical p-value is $1/201 \approx 0.005$. Wiener's two-descriptor model [51] uses the path number W and the polarity number, which on trees is linear in the degree-pair counts. Without tuning it gives a higher point estimate than ridge on the decanes (Supplementary Section S11), so distance information beyond the counts may matter for boiling points within one carbon number.

Under the stated boxes and candidate budgets, ridge regression predicts the boiling points more accurately than every single-index model evaluated, with or without γ, in point estimate, though not more accurately than Wiener's model on the decanes. The margin is large on the decanes and small on the pooled set, where size alone gives $Q^2 = 0.938$, and no interval for these differences is reported here. Part of this lead reflects capacity: ridge fits one coefficient per count, the centred count matrix has rank 7 on the decanes and rank 8 on the pooled set (Section 4.1), and each index model fits one descriptor (with n_C in Test B). These conclusions concern acyclic alkanes.

7 Concluding remarks

For graphs of equal order, that the pure-imbalance points at nonzero algebraic γ discriminate the histogram of edge imbalances and nothing finer is Rada's theorem [37, Theorem 3.4], a limit that binds every index de-

pending on an edge only through its degree ratio [37, Proposition 2.5]. The contribution here is to locate where this limit binds. For trees the histogram carries the full degree-pair information up to order twelve and loses it at order thirteen, and for chemical trees at sixteen, with explicit colliding families at every larger order. The minimality of these orders rests on exhaustive enumeration. Along the imbalance axis two chemical graphs exchange order at most five times. Off the ratio plane γ only reweights the same counts, which can restore distinctions that the parent loses at particular rational exponents (Section 4.2) and never destroys one at algebraic $\gamma \neq 0$ and rational exponents.

Under random search at matched budgets, no predictive gain from the imbalance parameter was demonstrated. A gain on vaporisation enthalpies met the seed-count criterion of a pre-registered protocol, with a bootstrap interval that included zero. It shrank to about zero in an exploratory comparison once the parent was searched over a box that did not truncate it. Pre-registered tests on measured boiling points, a property on which γ had given mixed results, were negative. On the decanes the point estimate favoured the parent, with an interval that included zero and a selection-aware range too wide to support or exclude gains of practical size. There a linear model on the degree-pair counts had higher point estimates than every single index, and Wiener's two-descriptor model higher still.

The results of Sections 3–5 are exact. Their limits, and those of Section 6, are of four kinds.

(i) *Enumeration scope.* The minimality of the collision orders in Theorem 2(iii), equality in the identifiability bound of Supplementary Section S10 for $\Delta \leq 6$, and the recovery of discrimination off Π in Section 4.2 rest on exhaustive enumeration. No general argument is given.

(ii) *Theoretical scope.* Whether two graphs with $\Delta \leq 4$ can exchange order five times along γ is open (Section 3.1).

(iii) *Data quality.* The data sets are small and acyclic: 18 octanes, 35 nonanes (34 for ΔH_{vap}), 34 decanes and 100 pooled C6–C10 alkanes. Seven octane entropies and the compiled acentric factors

rest on secondary sources, and the vaporisation enthalpy of 2,2,3,3-tetramethylbutane has an unresolved reference state (Supplementary Section S8).

(iv) *Inferential scope.* The negative verdicts mean that no gain was demonstrated. They do not show that γ never helps, and no equivalence margin was registered (Section 6.4). Changes made after outcomes had been seen are listed in Supplementary Table S8. None changes a pre-registered verdict as reported, but the wide-box analysis, in which the nonane gain shrinks to about zero, is itself not pre-registered.

The problem closest to these results is extremal: to determine the extremal graphs for general parameter triples, beginning with the maximal chemical trees of the pure-imbalance points, which Section 5 leaves open.

Data availability. The supplementary sections and tables cited here are in the file `LOYOLA_supplement.pdf` of the reproducibility package [33], which also holds the data, the scripts and verifiers that recompute every tabulated value, and the nonane and boiling-point protocols, committed before the corresponding models were fitted. The NIST WebBook pages used are not redistributed; a script of the package downloads them and checks each against the copy from which the data sets were built.

Only part of the mathematics is machine-checked. Lean 4 with Mathlib checks the termwise bounds of Supplementary Section S10 but not their equality cases. It also checks the convexity of $\log LO$, the spanning statement of Lemma 1 and the order-thirteen profile collision, but not the transcendence, Laguerre and enumeration arguments (Supplementary Section S7; the file `LEAN_VERIFICATION.md` maps each statement to its Lean theorem).

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